Slutsky's Theorem

2026-04-17

Introduction

Slutsky's theorem is one of the workhorse results of asymptotic statistics. It says that we can substitute a consistent estimator for a constant in an asymptotic statement, without changing the limiting distribution. This simple-sounding result is what makes the Wald statistic have an asymptotic standard-normal distribution, what justifies using the sample SD in place of the population SD in a t-statistic's denominator, and what underlies countless routine asymptotic arguments in applied work.

Prerequisites

The reader should understand convergence in probability and convergence in distribution.

Theory

Slutsky's theorem. Suppose $Y_n \xrightarrow{d} Y$ and $Z_n \xrightarrow{P} c$, where c is a constant. Then:

- $Y_n + Z_n \xrightarrow{d} Y + c$,
- $Y_n Z_n \xrightarrow{d} cY$,
- $Y_n / Z_n \xrightarrow{d} Y/c$ (provided $c \neq 0$).

The key requirement is that Z_n converges to a *constant*, not to another random variable. Convergence of Z_n to a non-degenerate random variable does not generally justify these manipulations.

Typical use. Consider the Wald statistic for a mean:

$$W_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{s_n}.$$

We know $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0,1)$ by the CLT, and $s_n \xrightarrow{P} \sigma$ by the LLN applied to the sample variance. Writing

$$W_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \cdot \frac{\sigma}{s_n},$$

Slutsky's theorem gives $W_n \xrightarrow{d} \mathcal{N}(0,1) \cdot 1 = \mathcal{N}(0,1)$.

This is why the Wald statistic with an estimated SE still has a standard-normal limit. The same principle applies to Wald chi-squared statistics, score statistics, and almost every test statistic in asymptotic theory.

Assumptions

Slutsky's theorem requires (i) Z_n converges in probability to a constant, not to a random variable, and (ii) the combined operation (sum, product, quotient) is well-defined. For quotients the constant must not be zero; dividing by a small random denominator can still introduce tail issues at finite n that Slutsky does not warn about.

R Implementation

```r
set.seed(2026)

n <- 500
mu <- 3; sigma <- 2
reps <- 5000

wald_stats <- replicate(reps, {
  x <- rnorm(n, mu, sigma)
  sqrt(n) * (mean(x) - mu) / sd(x)
})

qqnorm(wald_stats, pch = ".")
qqline(wald_stats, col = "red")

c(mean = mean(wald_stats),
  sd   = sd(wald_stats))
```

We simulate the Wald statistic across 5000 replications. Slutsky's theorem predicts the limiting distribution is standard normal; the simulation confirms empirically.

Output & Results

```
    mean       sd
 0.00411   0.9997
```

The empirical mean is near zero and SD near one, confirming standard-normal behaviour. The Q-Q plot falls on the reference line.

Interpretation

Slutsky's theorem is rarely cited by name in applied papers, but its conclusions appear constantly. The phrase "by the CLT, and using the consistency of $\hat{\sigma}$ for σ, the Wald statistic is asymptotically standard normal" is a Slutsky argument compressed into a single sentence.

Practical Tips

- When you see an asymptotic argument that swaps an estimator for a parameter, that is (usually) Slutsky at work.
- Slutsky's theorem does not help when both sequences are non-degenerate; for that, joint convergence and the continuous mapping theorem are needed.
- In finite samples, the Slutsky approximation may be crude. For small n, the t-distribution (which accounts for the estimated denominator more precisely) is preferred over the Wald normal.
- Extensions hold for continuous functions: if $Y_n \xrightarrow{d} Y$ and $Z_n \xrightarrow{P} c$, then $g(Y_n, Z_n) \xrightarrow{d} g(Y, c)$ for continuous g at (y, c).
- The theorem is sometimes stated with Z_n taking values in any fixed dimensional space – useful when the "constant" is a vector or matrix (e.g., an estimated covariance).

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**

- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr